Riemann Sums: Problems 35–48

Step-by-step solutions using Definition 5.4.3

$$A = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x \quad \text{where} \quad \Delta x = \frac{b-a}{n}$$

Problems 35–40: Right Endpoints

$x_k^* = a + k\,\Delta x$

35 $f(x) = \frac{x}{2}$ over $[1, 4]$

Parameters

$$a = 1,\ b = 4,\ \Delta x = \frac{4-1}{n} = \frac{3}{n}$$

Right endpoint

$$x_k^* = 1 + \frac{3k}{n}$$

Function value

$$f(x_k^*) = \frac{1}{2}\left(1 + \frac{3k}{n}\right) = \frac{1}{2} + \frac{3k}{2n}$$

Riemann sum

$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[\frac{1}{2}\sum_{k=1}^{n} 1 + \frac{3}{2n}\sum_{k=1}^{n} k\right]$$

$$= \frac{3}{n}\left[\frac{n}{2} + \frac{3(n+1)}{4}\right] = \frac{15n+9}{4n}$$

Area

$$A = \lim_{n\to\infty} \frac{15n+9}{4n} = \frac{15}{4}$$

36 $f(x) = 5 - x$ over $[0, 5]$

Parameters	$a = 0,\ b = 5,\ \Delta x = \frac{5}{n}$
Right endpoint	$x_k^* = \frac{5k}{n}$
Function value	$f(x_k^*) = 5 - \frac{5k}{n}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{5}{n}\left[5n - \frac{5}{n} \cdot \frac{n(n+1)}{2}\right]$ $= \frac{5}{n}\left[\frac{5n-5}{2}\right] = \frac{25(n-1)}{2n}$
Area	$A = \lim_{n\to\infty} \frac{25n-25}{2n} = \frac{25}{2}$

37 $f(x) = 9 - x^2$ over $[0, 3]$

Parameters	$a = 0,\ b = 3,\ \Delta x = \frac{3}{n}$
Right endpoint	$x_k^* = \frac{3k}{n}$
Function value	$f(x_k^*) = 9 - \frac{9k^2}{n^2}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[9n - \frac{9}{n^2} \cdot \frac{n(n+1)(2n+1)}{6}\right]$ $= 27 - 9 - \frac{27}{2n} - \frac{9}{2n^2} = 18 - \frac{27}{2n} - \frac{9}{2n^2}$
Area	$A = \lim_{n\to\infty} \left(18 - \frac{27}{2n} - \frac{9}{2n^2}\right) = 18$

38 $f(x) = 4 - \frac{1}{4}x^2$ over $[0, 3]$

Parameters	$a = 0,\ b = 3,\ \Delta x = \frac{3}{n}$
Right endpoint	$x_k^* = \frac{3k}{n}$
Function value	$f(x_k^*) = 4 - \frac{9k^2}{4n^2}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[4n - \frac{9}{4n^2} \cdot \frac{n(n+1)(2n+1)}{6}\right]$ $= 12 - \frac{9}{4} - \frac{27}{8n} - \frac{9}{8n^2} = \frac{39}{4} - \frac{27}{8n} - \frac{9}{8n^2}$
Area	$A = \lim_{n\to\infty}\left(\frac{39}{4} - \frac{27}{8n} - \frac{9}{8n^2}\right) = \frac{39}{4}$

39 $f(x) = x^3$ over $[2, 6]$

Parameters	$a = 2,\ b = 6,\ \Delta x = \frac{4}{n}$
Right endpoint	$x_k^* = 2 + \frac{4k}{n}$
Function value	$f(x_k^*) = 8 + \frac{48k}{n} + \frac{96k^2}{n^2} + \frac{64k^3}{n^3}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{4}{n}\left[8n + \frac{48}{n} \cdot \frac{n(n+1)}{2} + \frac{96}{n^2} \cdot \frac{n(n+1)(2n+1)}{6}\right.$ $\left. + \frac{64}{n^3} \cdot \frac{n^2(n+1)^2}{4}\right] = 320 + \frac{416}{n} + \frac{128}{n^2}$
Area	$A = \lim_{n\to\infty}\left(320 + \frac{416}{n} + \frac{128}{n^2}\right) = 320$

40 $f(x) = 1 - x^3$ over $[-3, -1]$

Parameters	$a = -3,\ b = -1,\ \Delta x = \frac{2}{n}$
Right endpoint	$x_k^* = -3 + \frac{2k}{n}$
Function value	$f(x_k^*) = 28 - \frac{54k}{n} + \frac{36k^2}{n^2} - \frac{8k^3}{n^3}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{2}{n}\left[28n - \frac{54}{n}\cdot\frac{n(n+1)}{2} + \frac{36}{n^2}\cdot\frac{n(n+1)(2n+1)}{6} - \frac{8}{n^3}\cdot\frac{n^2(n+1)^2}{4}\right] = 22 - \frac{26}{n} + \frac{8}{n^2}$
Area	$A = \lim_{n\to\infty}\left(22 - \frac{26}{n} + \frac{8}{n^2}\right) = 22$

Problems 41–44: Left Endpoints

$x_k^* = a + (k-1)\,\Delta x$

41 $f(x) = \frac{x}{2}$ over $[1, 4]$

Parameters	$a = 1,\ b = 4,\ \Delta x = \frac{3}{n}$
Left endpoint	$x_k^* = 1 + \frac{3(k-1)}{n}$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[\frac{n}{2} + \frac{3}{2n}\cdot\frac{(n-1)n}{2}\right] = \frac{15n-9}{4n}$
Area	$A = \lim_{n\to\infty}\frac{15n-9}{4n} = \frac{15}{4}$

42 $f(x) = 5 - x$ over $[0, 5]$

Parameters $a = 0,\ b = 5,\ \Delta x = \frac{5}{n}$

Left endpoint $x_k^* = \frac{5(k-1)}{n}$

Riemann sum
$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{5}{n}\left[5n - \frac{5}{n}\cdot\frac{(n-1)n}{2}\right] = \frac{25(n+1)}{2n}$$

Area
$$A = \lim_{n\to\infty} \frac{25n+25}{2n} = \frac{25}{2}$$

43 $f(x) = 9 - x^2$ over $[0, 3]$

Parameters $a = 0,\ b = 3,\ \Delta x = \frac{3}{n}$

Left endpoint $x_k^* = \frac{3(k-1)}{n}$

Riemann sum
$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[9n - \frac{9}{n^2}\cdot\frac{(n-1)n(2n-1)}{6}\right]$$
$$= 18 + \frac{27}{2n} - \frac{9}{2n^2}$$

Area
$$A = \lim_{n\to\infty}\left(18 + \frac{27}{2n} - \frac{9}{2n^2}\right) = 18$$

44 $f(x) = 4 - \frac{1}{4}x^2$ over $[0, 3]$

Parameters $a = 0,\ b = 3,\ \Delta x = \frac{3}{n}$

Left endpoint $x_k^* = \frac{3(k-1)}{n}$

Riemann sum
$$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{3}{n}\left[4n - \frac{9}{4n^2}\cdot\frac{(n-1)n(2n-1)}{6}\right]$$
$$= \frac{39}{4} + \frac{27}{8n} - \frac{9}{8n^2}$$

Area
$$A = \lim_{n\to\infty}\left(\frac{39}{4} + \frac{27}{8n} - \frac{9}{8n^2}\right) = \frac{39}{4}$$

Problems 45–48: Midpoints

$$x_k^* = a + \left(k - \tfrac{1}{2}\right)\Delta x$$

45 $f(x) = 2x$ over $[0, 4]$

Parameters	$a = 0,\ b = 4,\ \Delta x = \frac{4}{n}$
Midpoint	$x_k^* = \frac{4}{n}\left(k - \frac{1}{2}\right)$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{32}{n^2}\sum_{k=1}^{n}\left(k - \frac{1}{2}\right) = \frac{32}{n^2}\left(\frac{n^2}{2}\right) = 16$
Area	$A = \lim_{n\to\infty} 16 = 16$

46 $f(x) = 6 - x$ over $[1, 5]$

Parameters	$a = 1,\ b = 5,\ \Delta x = \frac{4}{n}$
Midpoint	$x_k^* = 1 + \frac{4}{n}\left(k - \frac{1}{2}\right)$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{4}{n}\left[5n - \frac{4}{n}\left(\frac{n^2}{2}\right)\right] = \frac{4}{n}(3n) = 12$
Area	$A = \lim_{n\to\infty} 12 = 12$

47 $f(x) = x^2$ over $[0, 1]$

Parameters	$a = 0,\ b = 1,\ \Delta x = \frac{1}{n}$
Midpoint	$x_k^* = \frac{1}{n}\left(k - \frac{1}{2}\right)$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{1}{n^3}\sum_{k=1}^{n}\left(k - \frac{1}{2}\right)^2 = \frac{1}{n^3}\left(\frac{n^3}{3} - \frac{n}{12}\right)$ $= \frac{1}{3} - \frac{1}{12n^2}$
Area	$A = \lim_{n\to\infty}\left(\frac{1}{3} - \frac{1}{12n^2}\right) = \frac{1}{3}$

48 $f(x) = x^2$ over $[-1, 1]$

Parameters	$a = -1,\ b = 1,\ \Delta x = \frac{2}{n}$
Midpoint	$x_k^* = -1 + \frac{2}{n}\left(k - \frac{1}{2}\right)$
Riemann sum	$\sum_{k=1}^{n} f(x_k^*)\Delta x = \frac{2}{n}\left[\frac{4}{n^2}\left(\frac{n^3}{3} - \frac{n}{12}\right) - 2n + n\right]$ $= \frac{2}{3} - \frac{2}{3n^2}$
Area	$A = \lim_{n\to\infty}\left(\frac{2}{3} - \frac{2}{3n^2}\right) = \frac{2}{3}$